

Without attaching any gateway to subnet in route tables, the subnet is considered as private.

Agenda:

Bastion Host - what, why, where

Load Balancer - what, why, where, how

We can't able to connect private ec2 directly.

Bastion Host: it's an EC2 with public IP, high security.

Why?

For entering into the private instance.

Bastion Host will work as a mediator.

Where?

When the private EC2 is there, there we are using Bastion Host.

DIFFERENCE BETWEEN PUBLIC EC2, BASTION HOST:

Common things:

1. Public IP.
2. Security groups.
3. Launching is same.

Difference:

1. Public EC2 is used to deploy the application but Bastion host is used to enter into private instance securely.
2. Public EC2 doesn't have high security Bastion host will provide high security to the private instance.

Keypair we are creating inside bastion host will guard the private instance.

VPC -- $> /16 \rightarrow 32-16=16 \rightarrow 2^{(power(16))}=65,536$

Subnet -- $> /24 \rightarrow 32 - 24=8 \rightarrow 2^{(power(8))}=256$

We don't need any IP's to Gateways. So In route table, we are taking 0.0.0.0/0

Connecting to putty:

1. Paste the public IP under hostname, check port number(22), connection type(SSH).
2. If u want to change letter size, window--> appearance --> change --> fontsize(14).
3. Connection --> 60(seconds between keepalives).
4. Double click SSH --> double click Auth --> single click Credentials --> 1st option(browse)--> select the location of ppk file.

Inside bastion host commands:

1. `sudo su` --> becoming as root user.
2. `nano <name of .pem file>` --> creating .pem file in ec2 without this we can't enter into private EC2.

3. `ls -la` --> checking the permissions for the files, folders in that path.
4. `chmod 700 <name of .pem file>` --> changing mode from read, write to read, write, execution for .pem file.
5. ssh command from private EC2 --> to enter into private EC2.

We don't give update command in bastion host because we are not deploying application there.

We will give update command in the instance where we want to deploy the application(here private EC2.)

Inside pvt EC2:

1. `sudo su` --> become as root user.
2. `yum update -y` --> yum(Yellowdog Update & Modified) is the package manager for linux OS, updating the server.
3. `yum install httpd -y` --> httpd is the web server for linux OS, installing the web server.
4. `systemctl status httpd` --> checking the status of httpd.
5. `systemctl start httpd` --> start httpd if it is inactive.
6. `systemctl enable httpd` --> enable httpd if it is disabled.
7. `which git` --> checking whether git is installed or not.
8. `yum install git -y` --> if git is not installed, install git.
9. `git clone <repo URL>` --> cloning the repo for deployment.
10. `ls` --> checking project folder is present or not.
11. `cd <project folder>` --> entering into the folder, cd(change directory).
12. Find html files.
13. `cp -r . /var/www/html` --> copying(cp) html files to destination path(/var/www/html), -r (recursive)copying all the files without giving the command multiple times.